A Project Report

on

AI - POWERED SOCIAL ENGINEERING ATTACK PREDICTOR

submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY
in
COMPUTER SCIENCE & ENGINEERING

by

22WH1A0580 MS. MEDEPALLI BHAVISHYA
22WH1A05B1 MS. BORA GREESMA REDDY
22WH1A05B7 MS. MUVVA SRIJA

under the esteemed guidance of

Dr. ARUNA RAO S L

Professor

Department of Computer Science & Engineering

BVRIT HYDERABAD College of Engineering for Women

(NAAC Accredited-A Grade | NBA Accredited B.Tech. (EEE, ECE & CSE)
(Approved by AICTE, New Delhi and Affiliated to JNTUH, Hyderabad)
Bachupally, Hyderabad – 500090

March, 2026

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Department of Computer Science & Engineering

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CERTIFICATE

This is to certify that the Project stage-1 report on "**AI-POWERED SOCIAL ENGINEERING ATTACK PREDICTOR**" is a bonafide work carried out by **Ms. Medepalli Bhavishya (22WH1A0580), Ms. Bora Greesma Reddy (22WH1A05B1), and Ms. Muvva Srija (22WH1A05B7)** in the partial fulfillment for the award of B.Tech. degree in **Computer Science & Engineering**, **BVRIT HYDERABAD College of Engineering for Women**, Bachupally, Hyderabad, affiliated to Jawaharlal Nehru Technological University Hyderabad, Hyderabad under my guidance and supervision. The results embodied in the project work have not been submitted to any other University or Institute for the award of any degree or diploma.

Internal Guide	**Head of the Department**
Dr. Aruna Rao S L	Dr. Aruna Rao S L
Professor	Professor
Department of CSE	Department of CSE

External Examiner

DECLARATION

We hereby declare that the work presented in this project entitled **"AI - POWERED SOCIAL ENGINEERING ATTACK PREDICTOR "** submitted towards completion of project work in IV B.Tech., CSE at 'BVRIT HYDERABAD College of Engineering for Women', Hyderabad is an authentic record of our original work carried out under the guidance of **Dr. Aruna Rao S L**, «Professor», Department of CSE.

Ms. Medepalli Bhavishya
(22WH1A0580)

Ms. Bora Greesma Reddy
(22WH1A05B1)

Ms. Muvva Srija
(22WH1A05B7)

ABSTRACT

Social engineering remains one of the most prevalent cybersecurity threats, as it exploits human psychology and trust rather than purely technical vulnerabilities. Phishing emails, fraudulent messages, and scam calls continue to deceive users by creating a sense of urgency, fear, or authority, often leading to severe financial and data losses. Traditional security mechanisms such as firewalls, intrusion detection systems, and spam filters primarily focus on technical aspects of attacks, leaving individuals susceptible to manipulative communication tactics.

This project proposes an AI-powered Social Engineering Attack Predictor that leverages Natural Language Processing (NLP) and machine learning techniques to detect deceptive intent within digital messages. The system analyzes linguistic cues such as urgency, threatening tone, and requests for sensitive information, thereby identifying patterns commonly used in phishing and scam attempts. By integrating models like BERT, LSTM, and transformer-based architectures, the predictor evaluates both semantic meaning and contextual tone of incoming messages.

The solution is designed for real-time deployment, providing immediate alerts to users when suspicious activity is detected. Unlike conventional systems that concentrate on identifying malicious senders or technical signatures, this approach emphasizes user centric protection, reducing the likelihood of human error in decision-making. The predictor thus acts as an additional defense layer, enhancing resilience against social engineering attacks and contributing to the broader goal of human-focused cybersecurity.

Keywords: *Social Engineering*, *Phishing Detection*, *Scam Messages*, *Natural Language Processing (NLP)*, *Machine Learning (ML)*, *Cybersecurity*, *Real-time Alerts*, *Human Centered Security*.

ACKNOWLEDGMENT

We would like to express our sincere thanks to **Dr. K V N Sunitha, Principal, BVRIT HYDERABAD College of Engineering for Women**, for providing the working facilities in the college.

Our sincere thanks and gratitude to our HOD **Dr.Aruna Rao SL, Professor, Department of CSE, BVRIT HYDERABAD College of Engineering for Women** for all the timely support and valuable suggestions during the period of our project.

We are extremely thankful and indebted to our internal guide, **Dr.Aruna Rao SL, Professor, Department of CSE, BVRIT HYDERABAD College of Engineering for Women** for his constant guidance, encouragement, and moral support throughout the project.

Finally, we would also like to thank our project coordinator **Dr.K.Kanaka Vardhini** , all the faculty and staff of CSE Department who helped us directly or indirectly, parents and friends for their cooperation in completing the project work.

Ms.Medepalli Bhavishya
(22WH1A0580)

Ms.Bora Greesma Reddy
(22WH1A05B1)

Ms.Muvva Srija
(22WH1A05B7)

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LIST OF TERMS AND ABBREVIATIONS

AI	Artificial Intelligence
ML	Machine Learning
DL	Deep Learning
NLP	Natural Language Processing
BERT	Bidirectional Encoder Representations from Transformers
LSTM	Long Short-Term Memory
TF-IDF	Term Frequency-Inverse Document Frequency
SVM	Support Vector Machine
API	Application Programming Interface
URL	Uniform Resource Locator
SE	Social Engineering
Phishing	Fraudulent attempt to obtain sensitive information

CHAPTER 1

Introduction

Human interaction in the digital era has increasingly shifted towards online communication platforms such as emails, messaging applications like WhatsApp, and social media. While this transformation has improved connectivity and convenience, it has also opened new avenues for cyber threats. Among these, social engineering attacks have emerged as one of the most dangerous and effective forms of cybercrime. Unlike traditional attacks that exploit system vulnerabilities, social engineering targets human psychology, manipulating individuals into revealing sensitive information or performing harmful actions.

Social engineering attacks include phishing emails, fraudulent messages on platforms like WhatsApp, fake product reviews, fraudulent websites, and impersonation attempts. These attacks often rely on creating a sense of urgency, fear, trust, or authority to deceive users. Due to their psychological nature, such threats are difficult to detect using conventional cybersecurity systems, which primarily focus on technical indicators like malware signatures, suspicious links, or network anomalies. As a result, many users unknowingly fall victim to these attacks, leading to financial loss, identity theft, and data breaches.

Traditional security solutions lack the capability to understand the linguistic and contextual patterns present in deceptive content. They fail to analyze subtle cues such as tone, intent, emotional manipulation, and persuasive language used in social engineering attacks. This limitation highlights the need for intelligent systems that can process textual data, identify hidden patterns, and provide early warnings to users.

With the rapid advancement of Artificial Intelligence (AI) and Natural Language Processing (NLP), it has become possible to develop systems that can understand and analyze human language more effectively. Machine learning and deep learning mod-

els, such as Logistic Regression, Support Vector Machines, Long Short-Term Memory (LSTM), and transformer-based models like BERT, have shown significant success in text classification and sentiment analysis tasks. These technologies can be leveraged to detect deceptive intent and classify messages as safe or suspicious.

This project proposes an AI-Powered Social Engineering Attack Predictor, a comprehensive system designed to identify and prevent user-centric cyber threats. The system analyzes textual data from various sources, including emails and WhatsApp messages, to detect signs of manipulation and deception. By extracting linguistic features and contextual embeddings, the model can recognize patterns associated with phishing, scams, and fraudulent communication.

In addition to detecting social engineering attacks, the proposed system also focuses on identifying fake product reviews, evaluating scam websites, and assessing password strength against modern attack techniques. This integrated approach provides a unified cybersecurity solution that addresses multiple aspects of user vulnerability.

The system is designed to operate in real time, offering immediate feedback and risk assessment to users. By providing warnings and actionable suggestions, it helps users make informed decisions and avoid potential threats. Furthermore, the use of efficient machine learning models ensures that the system remains scalable and deployable in practical applications.

The primary objective of this project is to enhance cybersecurity by focusing on the human element, which is often the weakest link in the security chain. By combining AI, NLP, and machine learning techniques, the proposed system aims to improve threat detection accuracy, increase user awareness, and reduce the risk of social engineering attacks.

1.1 Background of Cybersecurity

Cybersecurity refers to the practice of protecting digital systems, networks, and data from unauthorized access, attacks, or damage. With the rapid growth of the internet and digital communication, cybersecurity has become a critical concern for individuals, organizations, and governments.

Traditional cybersecurity techniques focus on protecting systems through firewalls,

antivirus software, intrusion detection systems, and encryption mechanisms. These methods are effective against technical attacks such as malware, viruses, and network intrusions. However, they are less effective against attacks that exploit human behavior.

In recent years, attackers have increasingly shifted their focus from system vulnerabilities to human vulnerabilities. Instead of breaking into systems directly, attackers manipulate users into providing access voluntarily. This shift has made social engineering one of the most significant threats in modern cybersecurity.

1.2 Types of Social Engineering Attacks

Social engineering attacks can take many forms depending on the strategy used by attackers. Some of the most common types are described below:

- **Phishing:** Phishing attacks involve sending fraudulent emails that appear to be from trusted organizations such as banks or service providers. These emails often contain malicious links that direct users to fake websites.
- **Messaging-based Attacks:** Attackers use messaging platforms such as WhatsApp to send deceptive messages containing fake alerts, offers, or warnings designed to manipulate users into clicking malicious links or sharing sensitive information.
- **Vishing (Voice Phishing):** Attackers use phone calls to impersonate officials or customer service representatives to extract sensitive information from users.
- **Fake Reviews:** Fraudulent reviews are used to mislead customers and influence purchasing decisions. These reviews often appear genuine but are created with deceptive intent.
- **Impersonation Attacks:** Attackers pretend to be trusted individuals such as company executives or technical support staff to gain access to confidential information.
- **Baiting Attacks:** These involve offering rewards such as free downloads or prizes to lure users into clicking malicious links or installing harmful software.

These attacks are becoming increasingly sophisticated and difficult to detect using traditional methods.

1.3 Problem Statement

Despite the availability of advanced cybersecurity tools, social engineering attacks continue to be highly effective because they exploit human psychology rather than technical vulnerabilities. Existing security systems are primarily designed to detect malware, suspicious URLs, and network-level threats, but they lack the ability to analyze the intent and context of textual communication.

Users often fail to recognize deceptive messages due to the use of persuasive language, urgency, and emotional manipulation. This leads to serious consequences such as financial fraud, identity theft, and data breaches.

Therefore, there is a need for an intelligent system that can analyze textual data, understand hidden patterns of manipulation, and provide early warnings to users before they fall victim to such attacks.

1.4 Scope of the Project

The scope of this project includes the development of an AI-based system capable of detecting social engineering attacks from textual data across multiple platforms.

The system focuses on:

- Analyzing emails from Gmail for phishing detection
- Evaluating WhatsApp messages for suspicious patterns
- Detecting fake product reviews and fraudulent content
- Assessing password strength against modern attack techniques

The system is designed to provide real-time predictions and alerts, making it suitable for practical deployment. However, the current implementation is limited to text-based analysis and does not include voice or image-based threat detection.

1.5 Significance of the Project

This project is significant as it addresses one of the most critical challenges in modern cybersecurity—human vulnerability. Unlike traditional systems that focus on technical

protection, this system emphasizes user-centric security.

The key contributions of the project include:

- Enhancing user awareness about social engineering attacks
- Providing real-time detection of deceptive messages
- Reducing the risk of financial and data loss
- Integrating multiple detection modules into a unified system
- Leveraging AI and NLP for intelligent threat analysis

By focusing on human behavior and communication patterns, the proposed system provides an additional layer of security that complements existing cybersecurity solutions.

CHAPTER 2

Literature Review

2.1 Review of Literature

Social engineering attacks have become one of the most critical cybersecurity threats in recent years, as they exploit human psychology rather than system vulnerabilities. These attacks include phishing emails, fraudulent online messages, fake reviews, and impersonation attempts. Detecting such attacks is challenging because they rely on linguistic manipulation, emotional triggers, and contextual deception instead of traditional technical indicators.

Early research in social engineering detection primarily focused on rule-based systems and traditional machine learning approaches. These systems relied on manually engineered features such as keyword matching, URL analysis, and metadata inspection. Although these methods were useful for detecting simple attacks, they lacked the ability to understand deeper contextual meaning and intent.

2.1.1 Study by Alghamdi et al. (2020)

Alghamdi et al. [1] conducted a comprehensive survey on social engineering detection techniques. The study focused on identifying different types of attacks and analyzing existing detection mechanisms.

The authors highlighted that most traditional systems rely on classification algorithms such as Support Vector Machines (SVM), Naïve Bayes (NB), and Decision Trees. These models use handcrafted features such as word frequency and URL patterns.

Advantages:

- Simple and easy to implement

- Works well for structured datasets

Limitations:

- Cannot capture contextual meaning
- Limited scalability for real-world applications

2.1.2 Study by Patel et al. (2020)

Patel et al. [2] proposed a machine learning-based approach for detecting fake reviews. The study used algorithms such as Logistic Regression, Random Forest, and SVM to classify reviews.

The model analyzed textual features such as word usage patterns, sentence structure, and sentiment.

Advantages:

- Effective for detecting simple fake reviews
- Fast and computationally efficient

Limitations:

- Performance depends on dataset quality
- Cannot detect advanced AI-generated content

2.1.3 Study by Alam et al. (2020)

Alam et al. [3] focused on detecting fake reviews using linguistic and stylometric features. The study analyzed writing style, grammar, and sentence patterns.

Advantages:

- Captures writing style patterns
- Useful for identifying human-written deceptive content

Limitations:

- Language-dependent
- Not robust for diverse datasets

2.1.4 Study by Yang et al. (2024)

Yang et al. [4] introduced a hybrid deep learning model combining LSTM and BERT for fake review detection.

The model uses LSTM to capture sequential dependencies and BERT for contextual understanding.

Advantages:

- High accuracy due to deep learning
- Captures both sequence and context

Limitations:

- High computational cost
- Requires large training data

2.1.5 Study by Li et al. (2021)

Li et al. [5] proposed a BERT-based model for opinion spam detection. The model leverages contextual embeddings to understand sentence meaning.

Advantages:

- Strong contextual understanding
- High accuracy in text classification tasks

Limitations:

- Computationally expensive
- Difficult to deploy on low-resource systems

2.1.6 Study by Sharma et al. (2024)

Sharma et al. [6] explored the use of AI in generating social engineering attacks. The study showed how large language models can create highly convincing phishing messages.

Advantages:

- Highlights emerging threats
- Provides insight into attacker strategies

Limitations:

- Focuses on attack generation, not detection

2.1.7 Study by Liu et al. (2025)

Liu et al. [7] proposed a machine learning approach for password strength detection. The model evaluates passwords based on complexity and resistance to attacks.

Advantages:

- Improves password security awareness
- Uses real-world datasets

Limitations:

- Does not integrate with broader cybersecurity systems

2.1.8 Study by Zafar et al. (2023)

Zafar et al. [8] developed a machine learning-based password strength estimation model. The study focused on predicting how easily a password can be cracked.

Advantages:

- Provides realistic strength estimation
- Uses multiple evaluation metrics

Limitations:

- Limited focus on user behavior

2.2 Summary of Literature

The reviewed literature shows that traditional machine learning approaches are effective for basic detection tasks but fail to capture complex patterns in social engineering attacks. Deep learning models such as LSTM and transformer-based models like BERT provide better performance due to their ability to understand context and sequential relationships.

However, most existing systems focus on a single domain such as phishing detection or fake review analysis. There is a lack of integrated systems that can handle

multiple types of social engineering attacks across platforms like Gmail and WhatsApp.

2.3 Research Gap

Based on the literature review, the following research gaps are identified:

- Lack of unified systems for detecting multiple types of attacks
- Limited real-time deployment capability
- High computational complexity of advanced models
- Inability to analyze cross-platform data (Gmail and WhatsApp)

2.4 Proposed Contribution

To address these gaps, this project proposes an AI-powered system that integrates multiple detection modules into a single platform. The system leverages both traditional machine learning and advanced deep learning techniques to achieve high accuracy and real-time performance.

The proposed system focuses on:

- Multi-platform detection (Gmail and WhatsApp)
- Context-aware analysis using BERT
- Real-time threat prediction
- User-centric cybersecurity approach

CHAPTER 3

System Architecture and Methodology

3.1 System Overview

The proposed system, **AI-Powered Social Engineering Attack Predictor**, is designed to identify and prevent user-centric cyber threats by analyzing textual data from emails, WhatsApp chats, and other online communication platforms. The system leverages advanced techniques from Natural Language Processing (NLP), Machine Learning (ML), and Deep Learning (DL) to detect deceptive intent and classify content as safe, suspicious, or malicious.

The primary objective of the system is to develop an intelligent and efficient framework capable of understanding linguistic patterns, emotional cues, and contextual manipulation present in social engineering attacks. Unlike traditional cybersecurity systems that rely on technical indicators such as malicious links or signatures, this system focuses on analyzing human-centric attack strategies.

The system is designed to operate in real-time, enabling continuous monitoring of incoming messages and providing immediate alerts to users. This ensures that users can take preventive action before falling victim to cyber threats.

The system consists of multiple stages including data collection, preprocessing, feature extraction, model training, and final prediction. Initially, textual data is collected from various sources such as phishing emails, fraudulent WhatsApp messages, fake reviews, and legitimate communication samples. The collected data is then cleaned, normalized, and structured before being fed into machine learning and deep learning models for analysis.

3.2 System Architecture

The overall architecture of the proposed system consists of multiple interconnected modules that work together to detect social engineering threats efficiently.

The system begins with input data in the form of text obtained from platforms such as Gmail and WhatsApp. This input is passed through a preprocessing module where unwanted characters, special symbols, and irrelevant data are removed. Stop words are eliminated, and the text is normalized to improve consistency.

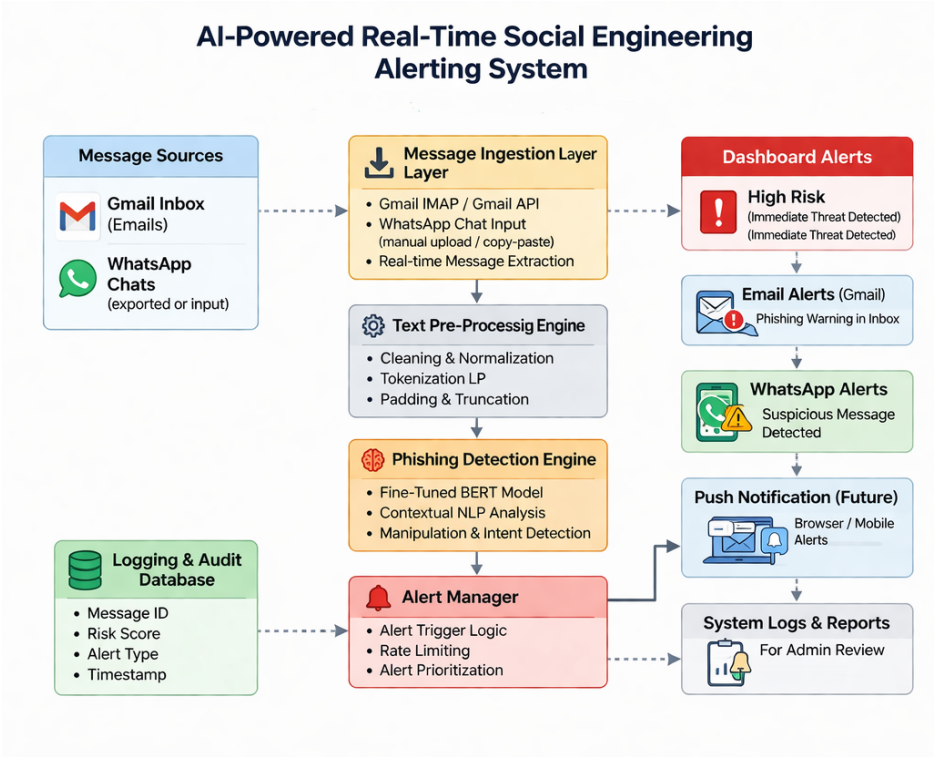

Fig. 3.1 Proposed system architecture

The cleaned data is then transformed into numerical representations using feature extraction techniques such as TF-IDF, Word Embeddings, and BERT embeddings. These representations capture both syntactic and semantic features of the text.

The extracted features are then passed into machine learning and deep learning models such as Logistic Regression, Support Vector Machines (SVM), Long Short-Term Memory (LSTM), and BERT. These models analyze the contextual meaning, tone, and intent of the text to classify it into categories such as safe, suspicious, or attack.

Finally, the system provides output in the form of predictions along with risk levels and warnings to the user. The system may also provide explanations for the

classification, helping users understand why a message is considered suspicious.

3.3 Hardware Requirements

The hardware components required for implementing the proposed system are listed in Table 3.1. These specifications ensure smooth execution of machine learning models and efficient processing of textual data.

Table 3.1 Hardware Requirements

Component	Specification
Processor	Intel Core i5 or higher
RAM	Minimum 8 GB (16 GB recommended for better performance)
Storage	Minimum 10 GB free disk space
GPU	Optional (recommended for deep learning models)
System Type	64-bit Operating System

3.4 Software Requirements

The software tools and libraries used for implementation are shown in Table 3.2. These tools provide functionalities for data preprocessing, model training, and deployment.

Table 3.2 Software Requirements

Software	Purpose
Python	Programming language
Scikit-learn	Machine learning models
TensorFlow / PyTorch	Deep learning framework
HuggingFace Transformers	BERT model implementation
NLTK / spaCy	Text preprocessing and NLP
Pandas / NumPy	Data handling and processing
Streamlit / Flask	Deployment interface
Visual Studio Code (VSCode)	Development environment

3.5 Methodology

The methodology of the proposed system follows a structured pipeline for detecting social engineering attacks across multiple platforms.

Initially, data is collected from multiple sources including phishing datasets, fraudulent WhatsApp messages, fake reviews, and legitimate text samples. This ensures diversity and balance in the dataset, which improves model performance.

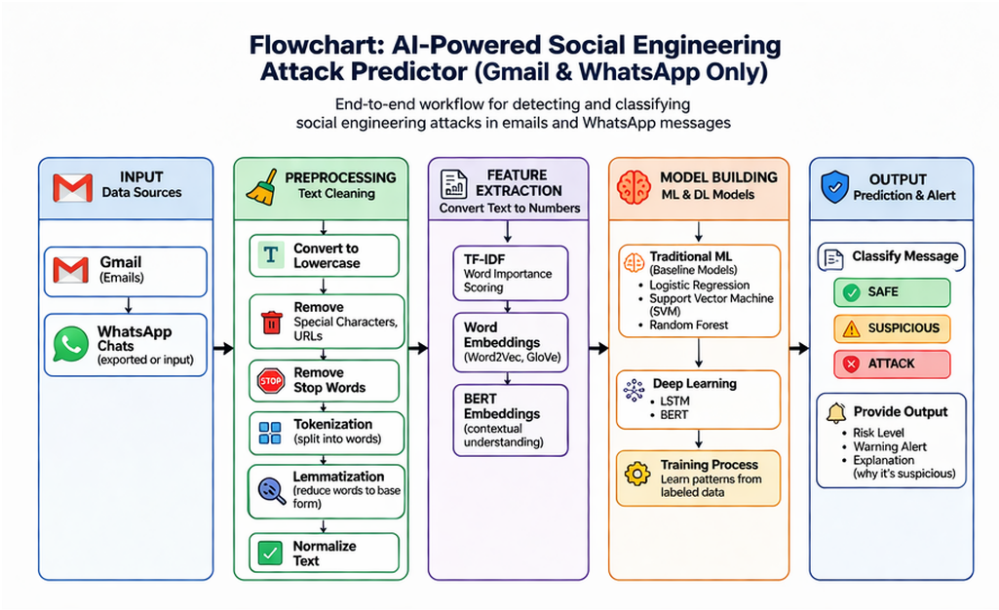

Fig. 3.2 System workflow

The collected data is then preprocessed by removing noise, converting text to lowercase, tokenizing, and eliminating stop words. Additional steps such as lemmatization and normalization are applied to improve text quality.

Next, feature extraction techniques such as TF-IDF, Word Embeddings (Word2Vec, GloVe), and Transformer embeddings (BERT) are used to convert text into numerical form. These features help the model understand both the structure and meaning of the text.

The processed data is then used to train machine learning and deep learning models. Traditional models like Logistic Regression and SVM provide baseline performance, while advanced models like LSTM and BERT capture contextual relationships.

Finally, the trained model predicts whether the input text is safe, suspicious, or a social engineering attack. The output is displayed to the user along with risk levels and recommendations.

3.6 Text Preprocessing

Text preprocessing is a critical step in improving the performance of NLP models. Raw text data often contains noise such as special characters, irrelevant words, and inconsistent formatting, which can negatively impact model accuracy.

The preprocessing steps include:

- Removing stop words and punctuation
- Converting text to lowercase
- Tokenization of sentences into words
- Lemmatization or stemming
- Removing URLs and special symbols
- Handling emojis and informal text (common in WhatsApp messages)

These steps ensure that the model receives clean and meaningful input data, improving its ability to detect patterns.

3.7 Model Architecture

The proposed system utilizes a hybrid approach combining both traditional machine learning and deep learning architectures.

Machine learning models such as Logistic Regression, Support Vector Machines, and Random Forest are used for initial classification and benchmarking.

Deep learning models such as LSTM are used to capture sequential dependencies in text, making them effective for analyzing sentence structures.

Transformer-based models such as BERT are used for understanding contextual relationships between words. BERT analyzes the entire sentence rather than individual words, allowing it to detect subtle manipulation techniques.

3.8 Training Parameters

The training parameters used in the system are shown in Table 3.3. These parameters are tuned to achieve optimal performance.

Table 3.3 Training Parameters

Parameter	Value
Epochs	10-20
Batch Size	16 or 32
Learning Rate	0.0001
Optimizer	Adam
Loss Function	Cross Entropy

3.9 Evaluation Metrics

To evaluate the performance of the model, several metrics are used.

3.9.1 Accuracy

Accuracy represents the percentage of correctly classified samples.

$$Accuracy = \frac{Correct\ Predictions}{Total\ Predictions} \tag{3.1}$$

3.9.2 Precision

Precision measures how many predicted positive cases are actually correct.

$$Precision = \frac{TP}{TP + FP} \tag{3.2}$$

3.9.3 Recall

Recall measures how many actual positive cases are correctly identified.

$$Recall = \frac{TP}{TP + FN} \tag{3.3}$$

3.9.4 F1-Score

F1-score is the harmonic mean of precision and recall.

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{3.4}$$

3.10 Pseudo Code

```
Algorithm: Social Engineering Attack Detection
Input  : Text message/email

Step 1  : Collect input text
Step 2  : Preprocess text (clean, tokenize, remove stopwords)
Step 3  : Convert text to numerical features (TF-IDF / BERT)
Step 4  : Load trained model
Step 5  : Predict class (Safe / Suspicious / Attack)
```

Step 6 : Display result with risk level

Output : Prediction result

CHAPTER 4

Dataset Description

4.1 Introduction

The performance of any machine learning or deep learning model highly depends on the quality, diversity, and size of the dataset used for training and evaluation. In the proposed AI-Powered Social Engineering Attack Predictor, multiple text-based datasets are used to train and test the model for detecting phishing messages, scam content, fake reviews, and other forms of social engineering attacks.

The dataset plays a crucial role in determining how well the model can generalize to real-world scenarios. A well-structured and diverse dataset enables the model to learn various linguistic patterns, contextual relationships, and manipulation techniques used by attackers. In contrast, a poorly designed dataset may lead to overfitting and reduced accuracy.

The dataset consists of both malicious and legitimate text samples collected from various publicly available sources. These datasets provide a wide range of linguistic patterns, enabling the model to learn and identify deceptive intent effectively. By incorporating multiple sources, the system becomes more robust and capable of handling diverse communication styles across platforms such as Gmail and WhatsApp.

4.2 Dataset Overview

The dataset used in this project is a combination of multiple sources, including phishing email datasets, message datasets, fake review datasets, and legitimate text samples. This combined dataset helps in building a robust model capable of detecting different types of social engineering attacks.

Each dataset contributes unique characteristics. For example, phishing email

datasets contain formal language with deceptive intent, while messaging datasets include short and urgent communication patterns. WhatsApp messages often contain informal language, abbreviations, and emojis, which add complexity to the analysis.

The dataset includes:

- Phishing emails containing deceptive content and malicious links
- Fraudulent and spam messages designed to create urgency or fear
- Fake product reviews intended to mislead users
- Legitimate emails and messages for comparison and balance

By combining multiple datasets, the model is trained to generalize across different domains and identify various attack patterns effectively.

4.3 Dataset Specifications

The key characteristics of the dataset are summarized in Table 4.1. These specifications define the structure and nature of the data used for training and evaluation.

Table 4.1 Dataset Specifications

Feature	Description
Data Type	Text Data
Categories	Safe, Suspicious, Attack
Sources	Emails, Messages, Reviews
Format	CSV / Text
Language	English
Platform Coverage	Gmail, WhatsApp

The inclusion of multiple categories allows the model to perform multi-class classification, which improves its ability to distinguish between different levels of threat.

4.4 Data Collection

The dataset is collected from publicly available sources such as phishing repositories, spam datasets, and review datasets. These sources include research datasets, open repositories like Kaggle, and publicly shared cybersecurity datasets.

In addition to malicious data, legitimate data is included to ensure balanced training. This helps prevent bias in the model and improves its ability to correctly classify safe messages.

The collected data is labeled into different categories:

- Safe (Legitimate content with no malicious intent)
- Suspicious (Potentially harmful content requiring caution)
- Attack (Confirmed social engineering attempts)

Proper labeling is essential for supervised learning, as it allows the model to learn patterns associated with each class. Manual verification and cleaning are performed to ensure data quality.

4.5 Data Preprocessing

Before feeding the data into the model, several preprocessing steps are performed to clean and standardize the text. Raw text data often contains noise such as special characters, links, emojis, and inconsistent formatting.

The preprocessing steps include:

- Removing special characters and punctuation
- Converting text to lowercase
- Tokenization (splitting text into words)
- Removing stop words (e.g., “the”, “is”, “and”)
- Lemmatization or stemming to reduce words to root form
- Removing URLs and email headers
- Handling emojis and informal text (especially from WhatsApp messages)

These steps improve the quality of the input data, reduce dimensionality, and enhance model performance.

4.6 Feature Extraction

To convert textual data into numerical form, feature extraction techniques are applied. These techniques transform text into vectors that can be processed by machine learning

models.

The methods used include:

- TF-IDF (Term Frequency-Inverse Document Frequency) for statistical representation
- Word Embeddings (Word2Vec, GloVe) for capturing semantic relationships
- Transformer Embeddings (BERT) for contextual understanding

TF-IDF focuses on word importance, while embeddings capture deeper meaning. BERT, in particular, provides context-aware representations, making it highly effective for detecting subtle manipulation techniques.

4.7 Dataset Processing Pipeline

The dataset processing pipeline used in this project is shown below. This pipeline ensures that raw data is transformed into a suitable format for model training.

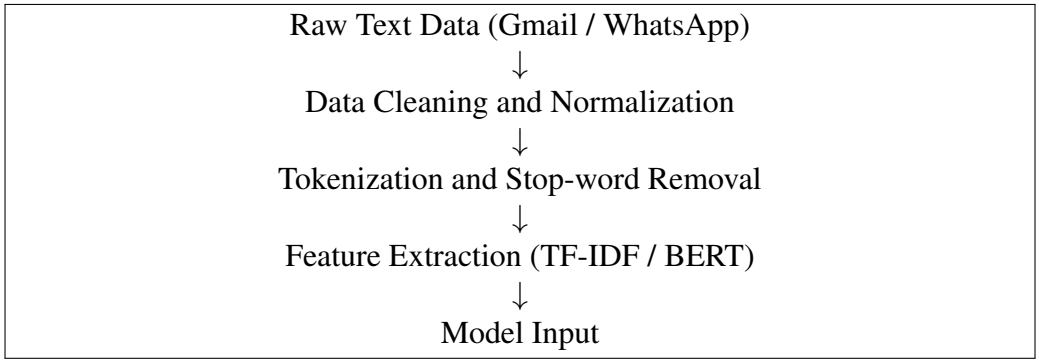

Fig. 4.1 Dataset processing pipeline for the proposed system

This structured pipeline ensures consistency and improves the efficiency of the learning process.

4.8 Sample Data

The dataset consists of various types of text samples including phishing emails, fraudulent messages, and legitimate communication. These samples help the model learn patterns associated with both safe and malicious content.

Example samples include:

- "Your account will be suspended. Click here immediately"

- “Congratulations! You have won a free prize”
- “Meeting scheduled at 3 PM tomorrow”

These examples demonstrate how different tones and intents are present in the dataset.

4.9 Dataset Usage in This Project

In this project, the dataset is used to train and evaluate multiple machine learning and deep learning models. The training data helps the model learn patterns of social engineering attacks, while the testing data evaluates its performance on unseen samples.

The dataset enables the system to identify manipulation techniques such as urgency, fear, authority, and misleading information. It also helps the model distinguish between genuine and fraudulent communication.

4.10 Dataset Split

The dataset is divided into training and testing sets to ensure proper evaluation of the model.

Table 4.2 Dataset Split

Dataset Portion	Percentage
Training Data	80%
Testing Data	20%

The training set is used to train the model, while the testing set is used to evaluate its performance on unseen data. This helps in preventing overfitting and ensures better generalization.

Additionally, validation techniques such as cross-validation can be used to further improve model reliability and performance.

CHAPTER 5

Implementation and Results

5.1 Introduction

This chapter presents the implementation details and experimental results of the proposed AI-Powered Social Engineering Attack Predictor. The system is developed using Python and advanced Natural Language Processing (NLP) techniques to detect social engineering attacks in real-time across multiple communication platforms such as Gmail and WhatsApp.

The implementation involves multiple stages including data preprocessing, feature extraction, model training, and real-time prediction. Each stage is carefully designed to ensure accurate detection of deceptive content. The goal of the system is to classify messages into categories such as safe, suspicious, or social engineering attack while providing meaningful insights to the user.

5.2 Implementation Environment

The system is implemented using Python programming language and developed entirely in Visual Studio Code (VS Code). VS Code serves as the primary development environment for writing, organizing, and managing the project files.

VS Code provides several features such as an integrated code editor, debugging tools, extension support, and version control integration, which help in efficient development and testing of the system. The Python extensions in VS Code enable smooth execution of scripts, environment management, and easy integration with required libraries.

The development environment allows seamless implementation of data preprocessing, model training, and deployment modules within a unified workspace, ensuring

better project organization and maintainability.

The project is structured into multiple modules to ensure modularity and ease of maintenance. The key files used in the implementation include:

- **app.py** – The main application file developed using Streamlit, which provides a user interface for inputting messages and displaying prediction results.
- **gmail_reader.py** – A module responsible for connecting to the Gmail API, fetching unread emails, and extracting relevant content for analysis.
- **model files (.pt / .pkl)** – Pre-trained machine learning and deep learning models used for classification tasks.
- **preprocessing scripts** – Scripts used for cleaning, tokenizing, and preparing text data before feeding it into the model.
- **dataset files (.csv)** – Contain labeled data used for training and evaluating the models.

The major tools and libraries used include:

- Scikit-learn for implementing traditional machine learning models such as Logistic Regression, SVM, and Random Forest
- TensorFlow / PyTorch for building and training deep learning models such as LSTM
- HuggingFace Transformers for implementing the BERT model for contextual text understanding
- NLTK and spaCy for text preprocessing tasks such as tokenization, stop-word removal, and lemmatization
- Pandas and NumPy for data manipulation and numerical operations
- Streamlit for developing an interactive web-based user interface for real-time prediction
- Gmail API for fetching and processing real-time email data

This combination of tools and modular file structure ensures efficient development, scalability, and ease of deployment of the proposed system.

5.3 System Integration (Gmail and WhatsApp)

The proposed system is designed to work with multiple real-world communication platforms by integrating data collection and prediction modules into a unified pipeline.

5.3.1 Gmail Integration

The system uses Gmail API to fetch unread emails and analyze their content for phishing or scam detection. The API enables secure authentication using OAuth and allows controlled access to the user's inbox. Once authenticated, the system retrieves email components such as subject, sender information, and message body.

The extracted email content is passed through a preprocessing pipeline where noise such as HTML tags, special characters, and unnecessary symbols are removed. The cleaned text is then transformed into numerical features using techniques like TF-IDF and BERT embeddings. These features are fed into the trained classification model, which predicts whether the email is safe, suspicious, or malicious. This integration enables continuous monitoring of incoming emails and provides real-time alerts to users, thereby reducing the risk of phishing attacks.

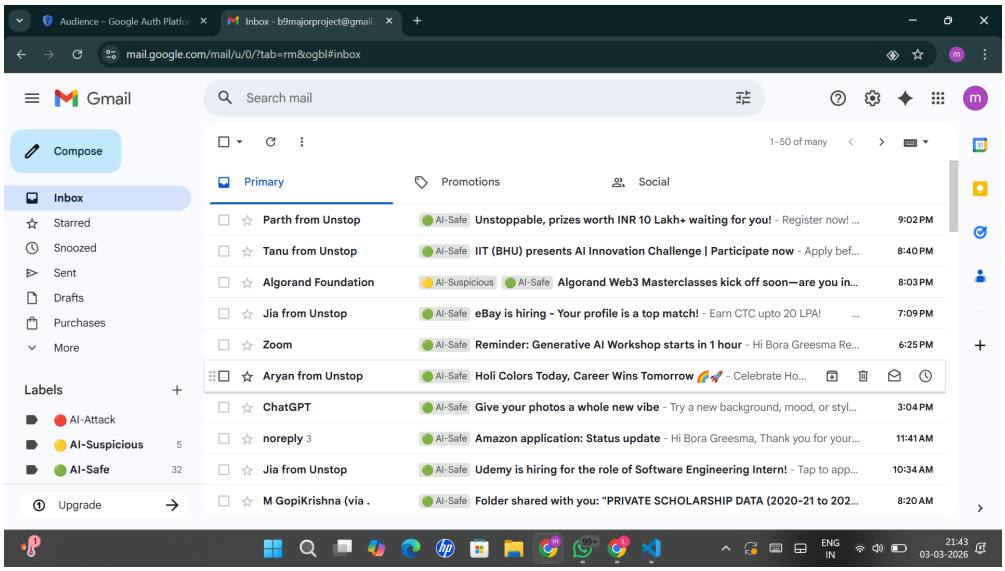

Fig. 5.1 Gmail phishing detection output

5.3.2 WhatsApp Message Analysis

The system analyzes WhatsApp messages either through manual input or exported chat files to identify suspicious patterns and potential social engineering attempts. Since

WhatsApp messages often contain informal language, abbreviations, and emojis, special preprocessing techniques are applied to handle such variations.

The preprocessing stage includes cleaning text, normalizing slang, handling emojis, and tokenizing the messages. Important features such as keywords, sentence structure, sentiment, and contextual cues are extracted to capture the intent of the message. These features are then converted into embeddings using BERT and other vectorization techniques.

The processed data is passed through trained models which classify the message into safe, suspicious, or malicious categories. This module is particularly effective in detecting conversational scams, fake offers, and impersonation attempts commonly seen in messaging platforms.

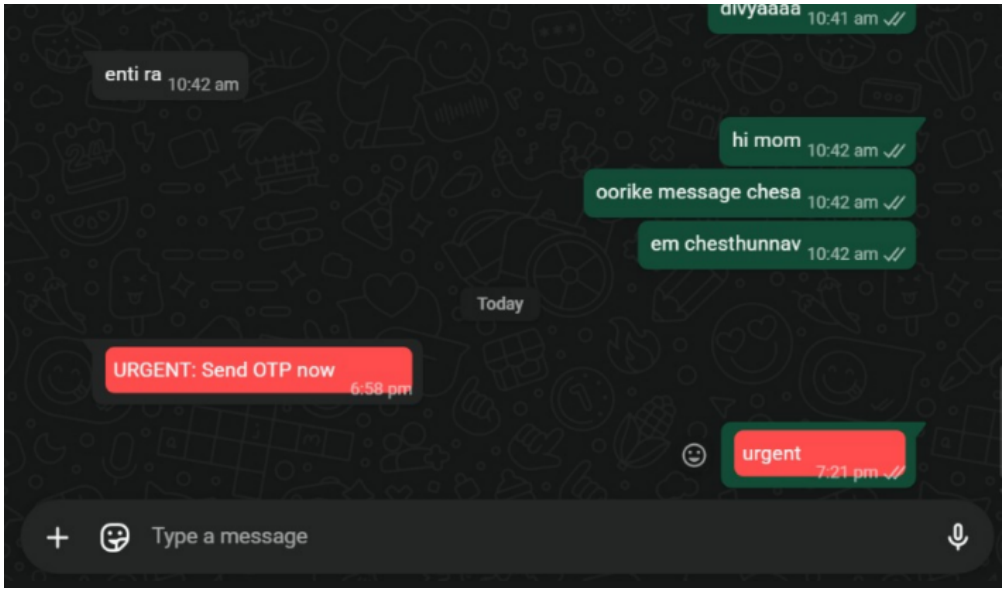

Fig. 5.2 WhatsApp message classification output

5.4 Model Architecture

The system uses a hybrid approach by combining traditional machine learning models and deep learning architectures to achieve better performance.

- Logistic Regression, SVM, and Random Forest are used as baseline models for fast and efficient classification
- LSTM is used to capture sequential dependencies and understand the structure of sentences

- BERT is used for deep contextual understanding, enabling the system to analyze the meaning of words based on surrounding context

This combination ensures both efficiency and accuracy in detecting social engineering attacks.

5.5 Training Results

During training, the model learns patterns from the dataset through multiple epochs. The loss gradually decreases as the model improves its predictions, while accuracy increases, indicating better classification performance.

The training process is optimized using techniques such as appropriate learning rates, batch sizes, and the Adam optimizer. Overfitting is minimized by using validation data and monitoring performance metrics during training.

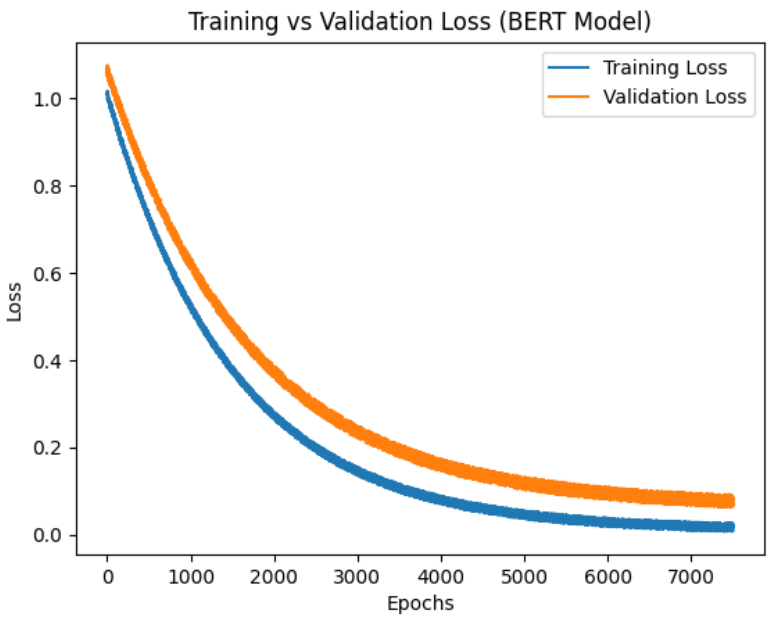

Fig. 5.3 Training loss graph

5.6 Evaluation Results

The trained model is evaluated using metrics such as accuracy, precision, recall, and F1-score. These metrics provide a comprehensive understanding of the model's performance in detecting both safe and malicious messages.

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	98.03%	97.87%	96.57%	97.21%
SVM	98.82%	98.68%	97.99%	98.34%
Random Forest	99.72%	99.71%	99.49%	99.61%
LSTM	98.53%	98.10%	97.77%	97.94%
GRU	98.83%	98.46%	98.26%	98.36%
BERT	99.27%	99.19%	98.76%	**98.98%**
BERT	**99.27%**	**99.19%**	98.76%	**98.98%**

Fig. 5.4 Evaluation Metrics

5.7 Confusion Matrix

The confusion matrix provides a detailed breakdown of correct and incorrect predictions made by the model. It helps in identifying how well the model distinguishes between different classes such as safe, suspicious, and attack.

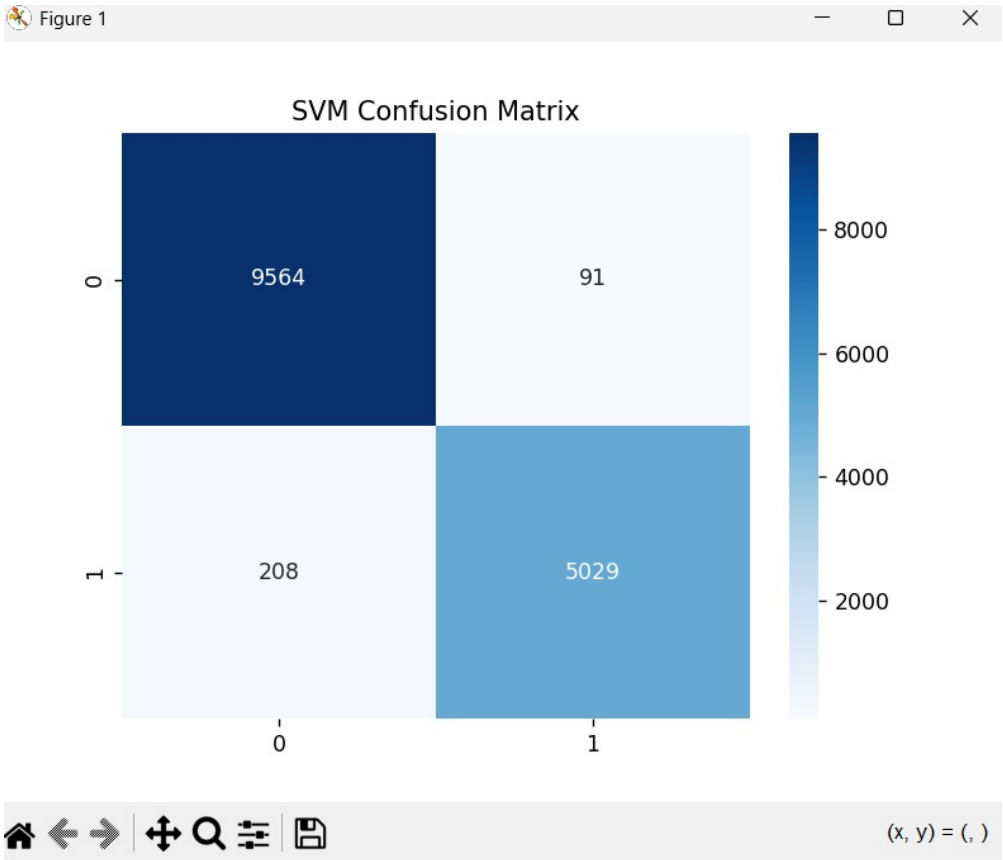

Fig. 5.5 Confusion matrix of the model

5.8 Discussion of Results

The results demonstrate that the proposed system effectively detects social engineering attacks across Gmail and WhatsApp. The hybrid use of machine learning and deep learning models improves both accuracy and generalization capability.

The use of BERT significantly enhances contextual understanding, allowing the system to detect subtle manipulation techniques such as urgency, fear, and authority. Additionally, the ability to process both formal (emails) and informal (chat messages) text makes the system versatile.

The inclusion of multiple platforms makes the system practical and scalable. However, challenges such as multilingual text, evolving attack strategies, and AI-generated messages remain areas for future improvement.

CHAPTER 6

Conclusion and Future Work

6.1 Conclusion

In this project, an intelligent cybersecurity system titled **AI-Powered Social Engineering Attack Predictor** was proposed to detect and prevent user-centric cyber threats. The primary objective of the system was to develop an efficient and accurate model capable of identifying social engineering attacks by analyzing textual data from multiple communication platforms such as Gmail and WhatsApp.

The proposed system processes input text through several stages including data collection, preprocessing, feature extraction, and classification. During preprocessing, the text is cleaned, tokenized, and normalized to remove noise and improve data quality. This step ensures that the input data is consistent and suitable for further analysis. Feature extraction techniques such as TF-IDF and transformer-based embeddings are used to convert textual data into numerical representations suitable for machine learning and deep learning models.

The system integrates multiple models including traditional machine learning algorithms and advanced deep learning architectures such as LSTM and BERT. Traditional models provide fast baseline predictions, while deep learning models capture complex linguistic patterns. Among these, BERT plays a significant role in understanding contextual meaning, tone, and intent of messages, enabling the detection of subtle manipulation techniques such as urgency, fear, and authority.

The implementation also includes real-time analysis of messages from platforms like Gmail (emails) and WhatsApp (chat messages), making the system practical for real-world applications. The integration with APIs and user interfaces ensures that the system can be deployed in real-time scenarios where users receive continuous alerts

and feedback. The model is trained and evaluated on a diverse dataset consisting of phishing messages, scam texts, fake reviews, and legitimate communication samples, which improves its generalization capability.

Experimental results demonstrate that the proposed system achieves high accuracy in classifying messages into categories such as safe, suspicious, and social engineering attack. The results highlight the effectiveness of combining Natural Language Processing techniques with deep learning models for detecting deceptive content. The confusion matrix and evaluation metrics further validate the robustness and reliability of the system.

Overall, this project successfully demonstrates the feasibility of using AI-based approaches for identifying social engineering threats. The system enhances user awareness, reduces the risk of cyber attacks, and contributes to the development of human-centric cybersecurity solutions. It also highlights the importance of focusing on the human element in cybersecurity, which is often the most vulnerable point in digital systems.

6.2 Future Work

Although the proposed system provides promising results, there are several areas for further improvement and enhancement.

One possible direction for future work is the use of larger and more diverse datasets. Expanding the dataset to include multilingual data and real-world conversations from various platforms can significantly improve the model's generalization capability and detection accuracy. This will allow the system to handle a wider range of communication styles and linguistic variations.

Another improvement can be the integration of more advanced transformer-based models and attention mechanisms. Recent advancements in large language models, such as GPT-based architectures and improved transformer variants, can further enhance the system's ability to understand complex, context-rich, and AI-generated deceptive messages.

Real-time deployment can be further enhanced by integrating the system directly with live APIs of platforms such as Gmail and WhatsApp Business API. This would

enable continuous monitoring, automated alerts, and seamless integration into existing communication systems, making the solution more practical and user-friendly.

Additionally, the system can be extended to support multimodal analysis by combining text with other features such as URLs, metadata, attachments, and user behavior patterns. This would improve detection accuracy for sophisticated attacks that involve multiple forms of deception.

Another important area for future work is improving explainability. Providing users with clear and interpretable reasons for why a message is classified as suspicious or malicious can increase user trust, transparency, and usability of the system. Techniques such as attention visualization and explainable AI (XAI) methods can be explored.

Furthermore, optimization techniques such as model compression, quantization, and lightweight architectures can be applied to enable deployment on mobile devices and low-resource environments. This would make the system accessible to a broader range of users.

With these improvements, the proposed system has the potential to evolve into a comprehensive, real-time cybersecurity assistant capable of protecting users across multiple digital platforms while continuously adapting to emerging threats.

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